MA1521 Calculus for Computing

Lecture Group 2

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Lesson 19: Tuesday 20 October 2026

(Week 10)

Remark. Q019.1 to Q19.6 are on Section 14.5 of Thomas' Calculus.

Q19.7 is on Section 14.8 on change of variable formula for triple integrals.



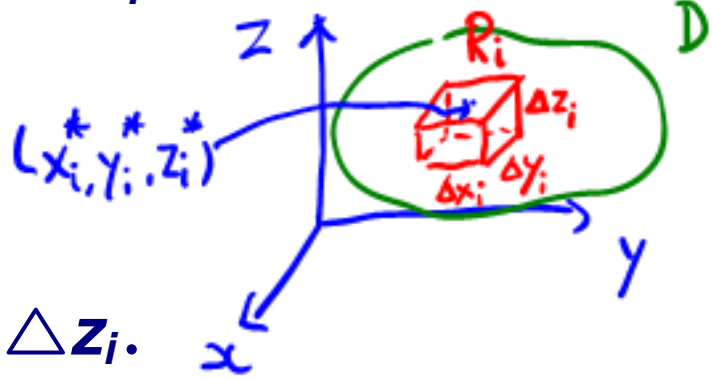
Triple Integrals

Suppose $f(x, y, z)$ is defined on a region D in xyz-space. We would mimic the construction of double integrals to define triple integrals $\iiint_D f(x, y, z)\, dV$ as follows:

Step 1. Divide D into n small rectangular solid regions R_i's as follows:

- Each small rectangular solid region R_i is of dimension $\triangle x_i$, $\triangle y_i$, $\triangle z_i$.
- So each R_i is of volume

$$\triangle V_i = \triangle x_i \triangle y_i \triangle z_i.$$



Step 2. Choose a point (x_i^*, y_i^*, z_i^*) in each R_i. Then for each R_i, we get an expression

$$f(x_i^*, y_i^*, z_i^*) \cdot \triangle V_i.$$

Now we take the Riemann sum:

$$\sum_{i=1}^{n} f(x_i^*, y_i^*, z_i^*) \cdot \triangle V_i. \qquad (1)$$

Step 3: Suppose $f(x, y, z)$ is continuous. When the subdivision is finer and finer, the Riemann sum in (1) will tend to a limit. We define

$$\iiint_D f(x, y, z)\, dV = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*, y_i^*, z_i^*) \cdot \triangle V_i.$$

Graph of $f(x,y,z)$: $w = f(x,y,z)$ solid in $xyzw$-"space"

Informally, we think of dV as the volume of a small solid subregion in D, so that we get a product $f(x, y, z)\, dV$ for the small solid region. Then the triple integral sign $\iiint_D$ means we take the limit of Riemann sum of these products $f(x, y, z)\, dV$.

Remark. If $f(x, y, z) \equiv 1$, then $\iiint_D 1\, dV$ is the volume of the solid region D.

In physics: density × volume = mass

Remark. A physical interpretation: Think of $f(x, y, z)$ as the density function of a solid D. Then $\iiint_D f(x, y, z)\, dV$ is the total mass of the solid D.

R = rectangular region in x-y plane

R: $a \le x \le b$, $c \le y \le d$

$$\iint_R f(x,y)\,dA = \int_{y=c}^{y=d}\int_{x=a}^{x=b} f(x,y)\,dx\,dy$$

Remark. Similar to double integrals, triple integrals can also be computed as iterated integrals.

$$= \int_{x=a}^{x=b}\int_{y=c}^{y=d} f(x,y)\,dy\,dx$$

For example, if $\boldsymbol{D}$ is rectangular solid region

$$\boldsymbol{a \le x \le b,\quad c \le y \le d,\quad e \le z \le f.}$$

Then

$$\iiint_D \boldsymbol{F(x,y,z)\,dV} = \int_{z=e}^{z=f}\int_{y=c}^{y=d}\int_{x=a}^{x=b} \boldsymbol{F(x,y,z)\,dx\,dy\,dz}$$

$$= \int_c^d \int_a^b \left(\int_e^f \boldsymbol{F(x,y,z)\,dz}\right)\boldsymbol{dx\,dy} = \cdots .$$

Q19.1 Evaluate the triple iterated integral

$$\int_{x=0}^{x=1}\int_{y=0}^{y=2}\left(\int_{z=0}^{z=3}(xy+2z)\,dz\right)dy\,dx.$$

innermost variable: z
middle variable: y
outermost variable: x

$$= \int_0^1\int_0^2 \left[xyz+z^2\right]_{z=0}^{z=3} dy\,dx$$

$$= \int_0^1\int_0^2 (3xy+9-(0+0))\,dy\,dx$$

$$= \int_0^1 \left(\int_0^2 (3xy+9)\,dy\right)dx$$

$$= \int_0^1 \left[\frac{3xy^2}{2}+9y\right]_{y=0}^{y=2} dx$$

$$= \int_0^1 (6x+18-(0+0))\,dx$$

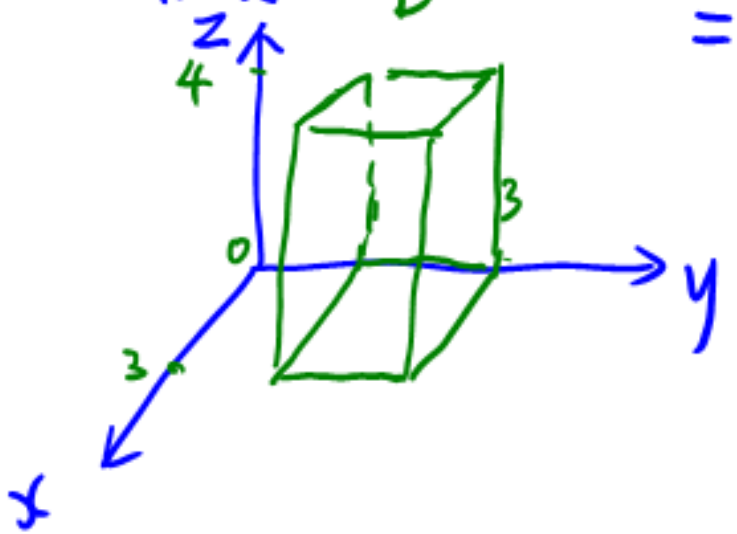
$$= \left[3x^2+18x\right]_0^1 = 3+18-(0+0) = 21.$$

Q19.2 Consider the region in the ***xyz***-space given by

$$D = \{(x, y, z) \ : \ 0 \le x \le 3, \ \ 1 \le y \le 3, \ \ 0 \le z \le 4\}.$$

Evaluate the triple integral $\iiint_D (x^2 + xyz)\, dV$ by computing an iterated integral in the order of ***dz dy dx***.

innermost variable: z
middle variable: y
outermost variable: x



$$\iiint_D (x^2+xyz)\, dV = \int_0^3 \int_1^3 \int_0^4 (x^2+xyz)\ dz\, dy\, dx$$

Solution:

$$\iiint_D (x^2 + xyz)\, dV$$

$$= \int_0^3 \int_1^3 \left(\int_0^4 (x^2 + xyz)\, dz \right) dy\, dx$$

$$= \int_0^3 \int_1^3 \left[x^2 z + \frac{1}{2} xyz^2 \right]_0^4 dy\, dx$$

$$= \int_0^3 \int_1^3 \left(4x^2 + \frac{1}{2} xy(4)^2 - (0 + 0) \right) dy\, dx$$

$$= \int_0^3 \left(\int_1^3 (4x^2 + 8xy)\, dy \right) dx$$

$$= \int_0^3 \left[4x^2 y + 4xy^2 \right]_1^3 dx$$

$$12x^2 + 36x - (4x^2 + 4x)$$

$$= \int_0^3 \left((4x^2 \cdot 3 + 4x \cdot 3^2) - (4x^2 \cdot 1 + 4x \cdot 1^2) \right) dx$$

$$= \int_0^3 \left((4x^2 \cdot 3 + 4x \cdot 3^2) - (4x^2 \cdot 1 + 4x \cdot 1^2)\right)\, dx$$

$$= \int_0^3 (12x^2 + 36x - 4x^2 - 4x)\, dx$$

$$= \int_0^3 (8x^2 + 32x)\, dx$$

$$= \left[\frac{8}{3}x^3 + 16x^2\right]_0^3$$

$$= \frac{8}{3} \cdot 3^3 + 16 \cdot 3^2 - 0$$

$$= 72 + 144 = 216.$$

Q19.3. Evaluate the triple integral in **Q19.2** again by computing an iterated integral in the order of ***dx dy dz***.

Solution: Alternatively, we have

$$\iiint_D (x^2 + xyz)\, dV$$

$$= \int_0^4 \int_1^3 \left(\int_0^3 (x^2 + xyz)\, dx \right) dy\, dz$$

$$= \int_0^4 \int_1^3 \left[\frac{x^3}{3} + \frac{x^2yz}{2} \right]_{x=0}^{x=3} dy\, dz$$

$$= \int_0^4 \int_1^3 \left(9 + \frac{9yz}{2} - (0+0) \right) dy\, dz$$

$$= \int_0^4 \left[9y + \frac{9z}{2} \cdot \frac{y^2}{2} \right]_{y=1}^{y=3} dz = \int_0^4 \left(\left(27 + \frac{81}{4}z\right) - \left(9 + \frac{9z}{4}\right) \right) dz$$

$$= \int_0^4 (18 + 18z)\, dz$$

$$= \int_0^4 \int_1^3 \left(9 + \frac{9}{2}yz\right)\, dy\, dz$$

$$= \int_0^4 \left[9y + \frac{9}{4}y^2 z\right]_1^3 \, dz$$

$$= \int_0^4 \left((9 \cdot 3 + \frac{9}{4} \cdot 3^2 z) - (9 \cdot 1 + \frac{9}{4} \cdot 1^2 z)\right)\, dz$$

$$= \int_0^4 \left(27 + \frac{81}{4}z - 9 - \frac{9}{4}z\right)\, dz$$

$$= \int_0^4 (18 + 18z)\, dz$$

$$= \left[18z + 9z^2\right]_0^4$$

$$= 18 \cdot 4 + 9 \cdot 4^2 - 0$$

$$= 72 + 144 = 216.$$

Suppose D is a solid region defined by:

$$D: \quad c \le y \le d, \quad g_1(y) \le x \le g_2(y), \quad h_1(x,y) \le z \le h_2(x,y).$$

Then

$$\iiint_D F(x,y,z)\,dV = \int_c^d \int_{g_1(y)}^{g_2(y)} \int_{h_1(x,y)}^{h_2(x,y)} F(x,y,z)\,dz\,dx\,dy,$$

etc.

In such case, the order of integration may or may not be reversed.

z: innermost variable (limits may depend on both middle & outermost variables)

x: middle variable (limits depend on outermost variable)

y: outermost variable (limits are numbers)

Q19.4. Consider the region in the ***xyz***-space given by

$$D = \{(x, y, z) : 0 \leq x \leq 1,\ 0 \leq y \leq 1 - x^2,\ 3 \leq z \leq 4 - x^2 - y\}.$$

Evaluate $\iiint_D x\, dV$.

Outermost variable: x
middle variable: y
innermost variable: z

$$= \int_0^1 \int_0^{1-x^2} \int_3^{4-x^2-y} x \, dz \, dy \, dx$$

Solution:

$$\iiint_D x\,dV$$

$$= \int_0^1 \int_0^{1-x^2} \left(\int_3^{4-x^2-y} x\,dz \right) dy\,dx$$

$$= \int_0^1 \int_0^{1-x^2} \left[xz \right]_{z=3}^{z=4-x^2-y} dy\,dx$$

$$= \int_0^1 \int_0^{1-x^2} \left(x(4-x^2-y) - 3x \right) dy\,dx$$

$$= \int_0^1 \left(\int_0^{1-x^2} \left(x - x^2 - xy \right) dy \right) dx$$

$$= \int_0^1 \left[xy - x^2 y - \frac{xy^2}{2} \right]_{y=0}^{y=1-x^2} dx$$

$$= \int_0^1 \left(x(1-x^2) - x^2(1-x^2) - \frac{x}{2}(1-x^2)^2 - (0+0+0) \right) dx$$

$$= \int_0^1 \left(x(1-x^2)^2 - \frac{x}{2}(1-x^2)^2 \right) dx$$

$$x(1-x^2) - x^2(1-x^2)$$

$$= (x - x^2)(1-x^2)$$

$$= x(1-x^2)^2$$

$$= \int_0^1 \left((x - x^3)(1 - x^2) - \frac{1}{2} x(1 - x^2)^2 - (0 - 0) \right) dx$$

$$= \int_0^1 \left(x(1 - x^2)^2 - \frac{1}{2} x(1 - x^2)^2 \right) dx$$

$$= \frac{1}{2} \int_0^1 x(1 - x^2)^2 \, dx = -\frac{1}{4} \int_0^1 (1 - x^2)^2 \cdot (-2x) \, dx$$

$$= -\frac{1}{4} \left[\frac{(1 - x^2)^3}{3} \right]_0^1$$

$$= -\frac{1}{4} \cdot (0 - \frac{1}{3}) = \frac{1}{12}.$$

Put $u = (1 - x^2)$

$\frac{du}{dx} = -2x$

$du = -2x \, dx$

$= -\frac{1}{4} \int (1 - u)^2 \, du$

Q19.5. Consider the solid region in the ***xyz***-space given by

$$D = \{(x, y, z) : 0 \le y \le 1,\ 0 \le x \le \sqrt{4 - y^2},\ 0 \le z \le 2x + y\}.$$

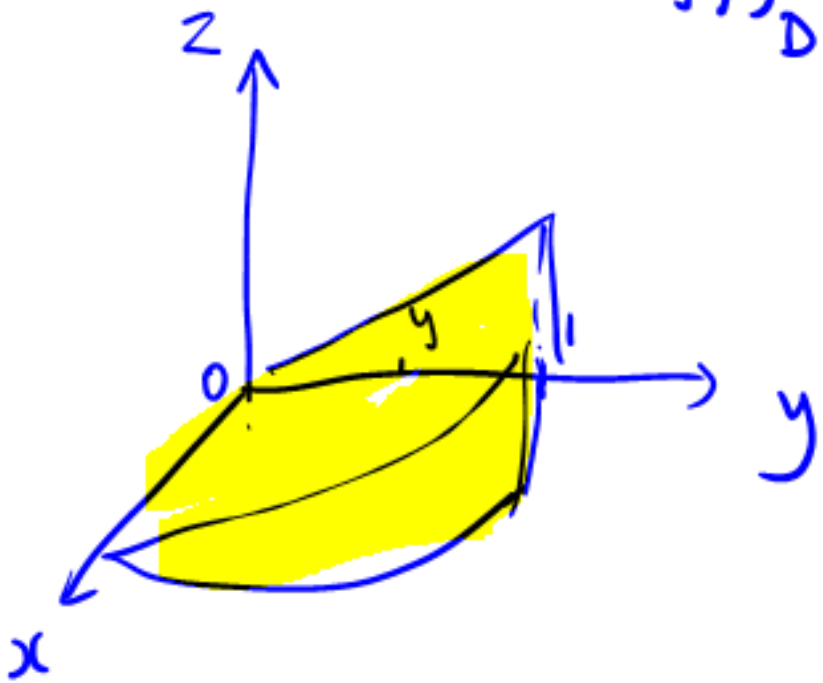
Find the volume of D.

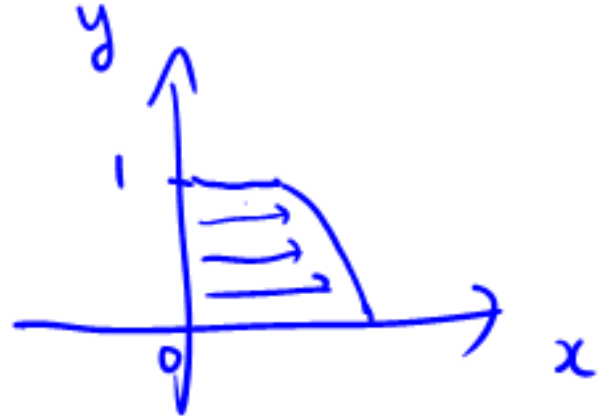
$$= \iiint_D 1 \, dV$$

outermost variable: y

middle variable: x

innermost variable: z





$x = \sqrt{4 - y^2}$

$x^2 = 4 - y^2$ $\quad x^2 + y^2 = 4$

$$\text{Volume} = \iiint_D 1 \, dV$$

$$= \int_0^1 \int_0^{\sqrt{4-y^2}} \left(\int_0^{2x+y} 1 \, dz \right) dx \, dy$$

$$= \int_0^1 \int_0^{\sqrt{4-y^2}} \Big[z \Big]_{z=0}^{z=2x+y} dx \, dy$$

$$= \int_0^1 \int_0^{\sqrt{4-y^2}} (2x+y-0) \, dx \, dy$$

$$= \int_0^1 \Big[x^2 + yx \Big]_{x=0}^{x=\sqrt{4-y^2}} dy$$

$$= \int_0^1 (4-y^2 + y\sqrt{4-y^2} - 0) \, dy$$

$$= \int_0^1 (4 - y^2 + y\sqrt{4 - y^2})\, dy$$

let $u = 4 - y^2$

$$= \left[4y - \frac{1}{3}y^3 - \frac{1}{3}(4 - y^2)^{3/2}\right]_0^1$$

$$= \left(4(1) - \frac{1}{3}(1)^3 - \frac{1}{3}(4 - 1^2)^{3/2}\right)$$

$$- \left(0 - 0 - \frac{1}{3}(4 - 0^2)^{3/2}\right)$$

$$= \left(4 - \frac{1}{3} - \sqrt{3}\right) + \frac{8}{3}$$

$$= \frac{11}{3} - \sqrt{3} + \frac{8}{3} = \frac{19}{3} - \sqrt{3}$$

$x \geq 0, y \geq 0, z \geq 0$

Q19.6. Consider the region W in the first octant of the xyz-space bounded by the coordinate planes $x = 0$, $y = 0$, $z = 0$, the plane $y = 1$ and the plane $x + z = 1$. Let $g(x, y, z)$ be a function defined on W.

(i) Write the triple integral $\iiint_W g(x, y, z)\, dV$ as an iterated integral with integration order $dz\, dx\, dy$.

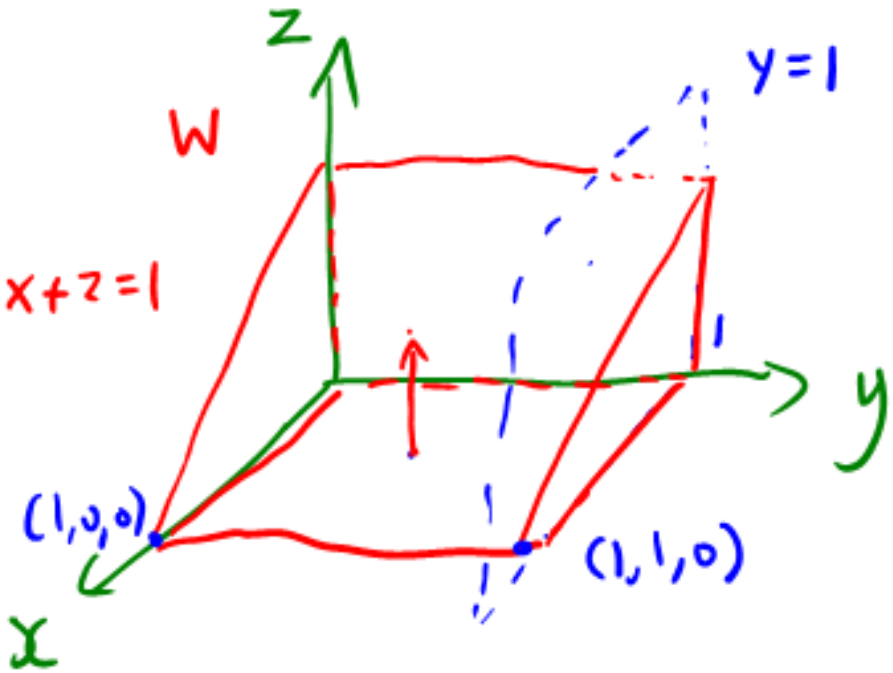
(ii) Write the triple integral $\iiint_W g(x, y, z)\, dV$ as an iterated integral with integration order $dy\, dx\, dz$.

Solution: (i) Objective: To write

$$\iiint_W g(x,y,z)\,dV = \int_a^b \int_c^d \int_e^f g(x,y,z)\,dz\,dx\,dy.$$

$W: \; a \le y \le b, \; c(y) \le x \le d(y), \; e(x,y) \le z \le f(x,y).$

$x=0$
$y=0$
$z=0$
$y=1$
$x+z=1$

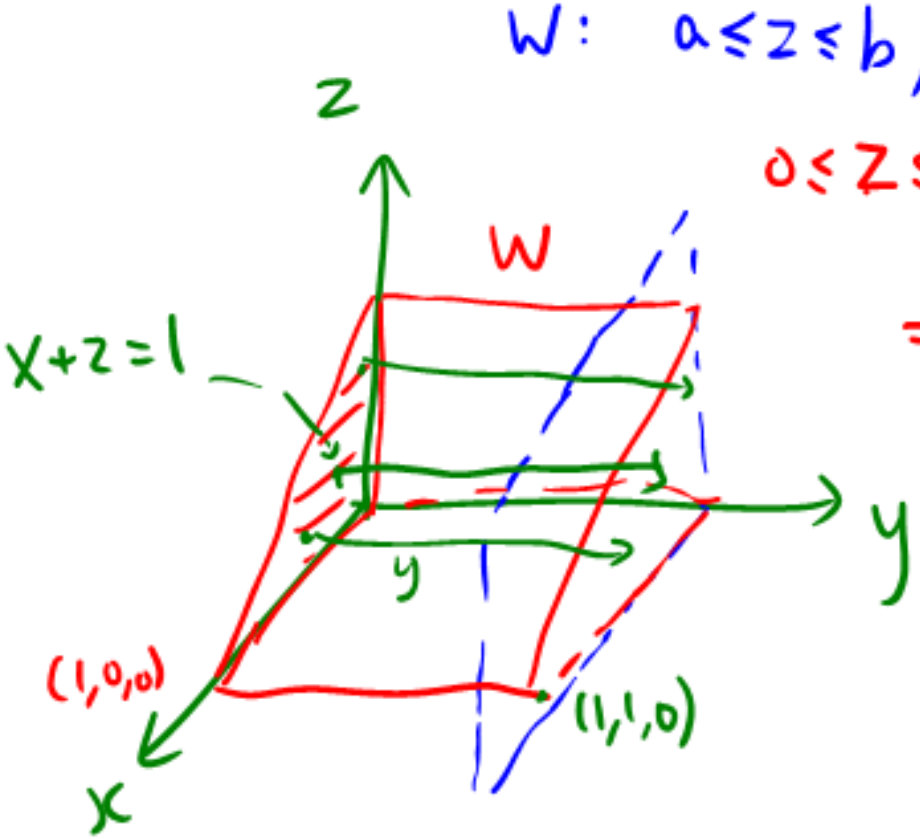


$$= \int_0^1 \int_0^1 \int_0^{1-x} g(x,y,z)\,dz\,dx\,dy$$

$W: \; 0 \le y \le 1, \; 0 \le x \le 1, \; 0 \le z \le 1-x$

Solution: (i) Objective: To write

$$\iiint_W g(x,y,z)\,dV = \int_a^b \int_c^d \int_e^f g(x,y,z)\,dy\,dx\,dz.$$



$0 \le z \le 1$, $0 \le x \le 1-z$, $0 \le y \le 1$

$$= \int_0^1 \int_0^{1-z} \int_0^1 g(x,y,z)\,dy\,dx\,dz$$

